Materials Design Analysis Reporting (MDAR) Checklist for Authors

The MDAR framework establishes a minimum set of requirements in transparent reporting applicable to studies in the life sciences (see Statement of Task: doi:10.31222/osf.oio/9sm4x.). The MDAR checklist is a tool for authors, editors and others seeking to adopt the MDAR framework for transparent reporting in manuscripts and other outputs. Please refer to the MDAR Elaboration Document for additional context for the MDAR framework.

Materials

Antibodies	Yes (indicate where provided: page no/section/legend)	n/a
For commercial reagents, provide supplier name, catalogue number and RRID, if available	No antibodes detected. Please add identifiers for all resources where possible	

Cell Materials	Yes (indicate where provided: page no/section/legend)	n/a
Cell lines: Provide species information, strain. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No cell lines detected Please add identifiers for all resources where possible	
Primary cultures: Provide species, strain, sex of origin, genetic modification status.	Not currently checked by SciScore	

Experimental Animals	Yes (indicate where provided: page no/section/legend)	n/a
Laboratory animals: Provide species, strain, sex, age, genetic modification status. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No organisms detected Please add identifiers for all resources where possible	
Animal observed in or captured from the field: Provide species, sex and age where possible	Not currently checked by SciScore	
Model organisms: Provide Accession number in repository (where relevant) OR RRID	See laboratory animals section for information.	

Plants and microbes	Yes (indicate where provided: page no/section/legend)	n/a
Plants: provide species and strain, unique accession number if available, and source (including location for collected wild specimens)	Not currently checked by SciScore	
Microbes: provide species and strain, unique accession number if available, and source	Not currently checked by SciScore	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
Identify authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	The study was approved by the Ethics Review Board, Uppsala (2018159).	
Provide statement confirming informed consent obtained from study participants.	Exclusion criteria were: celiac disease, functional dyspepsia, Helicobacter pylori infection during the preceding 6 mo, or other functional or inflammatory gastrointestinal disease, previous or ongoing cancer treatment, previous bariatric or abdominal surgery other than appendectomy, treatment of weight reduction, >10 kg body weight change in the preceding year, refusal to give informed consent, unstable medication from 14 d prior to inclusion or during the study, concurrent probiotic or antibiotic medication, reluctance to consume rice porridge daily during 3 separate weeks, pregnancy or lactation, blood donation or participation in other intervention trials within 30 d prior to screening or any time during the study, history of drug or alcohol abuse, smoking, and inability to understand the Swedish language.	
Report on age and sex for all study participants.	**Age:**Participants Inclusion criteria of the study were: female and male with moderate to severe IBS (IBS-SSS score >175) (28), BMI 18.5-38 kgm2, age 18-70 y, hemoglobin 120-160	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
	gL, thyroid-stimulating hormone <4 mUL, C-reactive protein <5 mgL, transglutaminase immunoglobulin A <7 UmL, and systolicdiastolic blood pressure <=160<=105 mmHg. **Sex:**Participants Inclusion criteria of the study were: female and male with moderate to severe IBS (IBS-SSS score >175) (28), BMI 18.5-38 kgm2, age 18-70 y, hemoglobin 120-160 gL, thyroid-stimulating hormone <4 mUL, C-reactive protein <5 mgL, transglutaminase immunoglobulin A <7 UmL, and systolicdiastolic blood pressure <=160<=105 mmHg.	

Design

Study protocol	Yes (indicate where provided: page no/section/legend)	n/a
For clinical trials, provide the trial registration number OR cite DOI in manuscript.	NCT03653689	

Laboratory protocol	Yes (indicate where provided: page no/section/legend)	n/a
Provide DOI or other citation details if detailed step-by-step protocols are available.	Not detected.	

Experimental study design (statistics details)	Yes (indicate where provided: page no/section/legend)	n/a
State whether and how the following have been done, or if they were not carried out		
Sample size determination	Randomization was restricted to three treatment orders , to ensure sufficient statistical power to allow differentiation of a potential effect of sequencing .	
Randomization	Methods Study design The double-blind, placebo-controlled, randomized 3-way study with triple crossover design was conducted from September 2018 to June 2019 in Uppsala, Sweden.	
Blinding	Participants were then prompted to consume blinded food in the form of rice porridge with added FODMAPs, gluten, or placebo during weeks 3, 5, and 7, but no porridge during weeks 4 and 6 (Figure 1).	
inclusion/exclusion criteria	Participants Inclusion criteria of the study were: female and male with moderate to severe IBS (IBS-SSS score >175) (28), BMI 18.5-38 kgm2, age 18-70 y, hemoglobin 120-160 gL, thyroid-stimulating hormone <4 mUL, C-reactive protein <5 mgL, transglutaminase immunoglobulin A <7 UmL, and systolicdiastolic blood pressure <=160<=105 mmHg.	

Sample definition and in-laboratory replication	Yes (indicate where provided: page no/section/legend)	n/a
State number of times the experiment was replicated in laboratory	Not detected.	
Define whether data describe technical or biological replicates	Not detected.	

Ethics	Yes (indicate where provided: page no/section/legend)	n/a
Studies involving human participants: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	The study was approved by the Ethics Review Board, Uppsala (2018159).	
Studies involving experimental animals: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Not detected.	
Studies involving specimen and field samples: State if relevant permits obtained, provide details of authority approving study; if none were required, explain why.	Procedures for the collection of blood, urine, and feces samples are described in detail in Supplementary Methods 2 .	

Dual Use Research of Concern (DURC)	Yes (indicate where provided: page no/section/legend)	n/a
If study is subject to dual use research of concern, state the authority granting approval and reference number for the regulatory approval	Not currently checked by SciScore	

Analysis

Attrition	Yes (indicate where provided: page no/section/legend)	n/a
State if sample or data point from the analysis is excluded, and whether the criteria for exclusion were determined and specified in advance.	Two individuals were mistakenly assigned the sequence ABC .	

Statistics	Yes (indicate where provided: page no/section/legend)	n/a
Describe statistical tests used and justify choice of tests.	Under these conditions, the required number of participants was 64, with the level of significance in a 2-sided test set to 0.053 (Bonferroni correction).	

Data availability	Yes (indicate where provided: page no/section/legend)	n/a
State whether newly created datasets are available, including protocols for access or restriction on access.	Not detected.	
If data are publicly available, provide accession number in repository or DOI or URL.	Not detected.	
If publicly available data are reused, provide accession number in repository or DOI or URL, where possible.	Not detected.	

Code availability	Yes (indicate where provided: page no/section/legend)	n/a
For all newly generated code and software essential for replicating the main findings of the study:		
State whether the code or software is available.	Not detected.	
If code is publicly available, provide accession number in repository, or DOI or URL.	Not detected.	

Analysis

Adherence to community standards	Yes (indicate where provided: page no/section/legend)	n/a
MDAR framework recommends adoption of discipline-specific guidelines, established and endorsed through community initiatives. Journals have their own policy about requiring specific guidelines and recommendations to complement MDAR.		
State if relevant guidelines (eg., ICMJE, MIBBI, ARRIVE) have been followed, and whether a checklist (eg., CONSORT, PRISMA, ARRIVE) is provided with the manuscript.	Not currently checked by SciScore	